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B.Tech Winter Semester 2024-25

School Of Computer Science and Engineering

(SCOPE)

Course Code and Name	BCSE414L & HIGHPERFORMANCE COMPUTING
Slot	A1+TA1
Register Number of the Student	22BCE5225
Name of the Student	S R KOUSALYA
Title of the Paper	ALZHEIMER'S PREDICTION OPTIMIZATION USING HPC
References (book/journal paper/web)	JOURNAL PAPER

1) ABSTRACT:

Alzheimer's disease prediction from MRI images involves processing large volumes of complex medical data, often leading to high computational demands. This research proposes a high-performance computing (HPC)-based deep learning approach to classify Alzheimer's stages. The system employs parallel processing, MPI-based distributed training, automated hyperparameter tuning, and CNN optimization, implemented via scripts such as parallel_processing.py and mpi_distributed_processing.py, benchmark_parallel_training.py, and cnn_training_hpc.py, predict_image.py, parallel_tuning_simulation.py. The model is trained on a labeled MRI dataset with four classes: Mild Demented, Very Mild Demented, Moderate Demented, and Non-Demented. Performance is evaluated using prediction accuracy, execution time, and confusion matrix analysis. Speedup efficiency, analysed through Amdahl's Law, offers insights into scalability and computational limits. Efficiency graphs further illustrate trade-offs between computational cost and performance. This integrated HPC framework delivers accurate, fast, and scalable predictions, enhancing early diagnosis of Alzheimer's disease.

2) LITERATURE REVIEW:

[1] Adaptive 3DCNN-Based Interpretable Ensemble Model for Early Diagnosis of Alzheimer's Disease

This paper presents an adaptive ensemble model, 3DCNN+EL+GA, combining 3D Convolutional Neural Networks and Genetic Algorithms for early Alzheimer's and MCI diagnosis. It outperforms existing deep learning methods on ADNI and OASIS datasets. The model identifies key brain regions contributing to classification using gradient-based attribution. These regions are linked to behavioral domains affected early in AD. The approach may also extend to diagnosing other brain disorders using neuroimaging data.

[2] Risk Prediction of Alzheimer's Disease Conversion in Mild Cognitive Impaired Population Based on Brain Age Estimation

This study proposes a Conversion Risk Estimation System (CRES) to predict Alzheimer's disease (AD) risk in Mild Cognitive Impairment (MCI) patients using MRI-based brain age estimation. CRES distinguishes between NC, sMCI, pMCI, and AD groups based on age gap (AG). A significant correlation is found: each extra AG year increases AD conversion risk by 4.57%. The model also incorporates gender and MMSE for accurate risk analysis. A nomogram provides personalized risk predictions up to 8 years from baseline.

[3] Leveraging Brain MRI for Biomedical Alzheimer's Disease Diagnosis Using Enhanced Manta Ray Foraging Optimization Based Deep Learning

This research introduces the ADD-EMRFODL model for Alzheimer's Disease (AD) diagnosis using brain MRI and deep learning. The method begins with Gabor Filtering (GF) for preprocessing, followed by feature extraction using DenseNet-121. A Back Propagation Neural Network (BPNN) is used for classification, with its parameters optimized through Enhanced Manta Ray Foraging Optimization (EMRFO). This eliminates the need for manual feature extraction and improves diagnostic efficiency. Experimental results demonstrate that the ADD-EMRFODL model achieves superior classification accuracy compared to existing techniques. The use of EMRFO for optimizing BPNN parameters further boosts classification accuracy, ensuring robust and reliable Alzheimer's diagnosis from MRI scans.

[4] Deep Learning-Based Diagnosis of Alzheimer's Disease Using Structural Magnetic Resonance Imaging: A Survey

This paper presents an Alzheimer's Disease (AD) is the most prevalent form of dementia, and early diagnosis is key to delaying its progression. Structural MRI (sMRI) is effective in detecting brain abnormalities associated with AD. With advancements in deep learning, numerous models have been applied to sMRI for AD detection with promising results. This paper reviews deep learning approaches categorized by input type, highlighting their strengths and limitations. It also discusses current challenges and proposes future research directions for improving AD diagnosis using deep learning.

[5] Graph-based deep learning models in the prediction of early-stage Alzheimer's

This study explores early-stage Alzheimer's prediction using graph-based deep learning models on rs-fMRI data. A comparative analysis of rs-fMRI time points and functional connectivity (FC) reveals FC's superior predictive value. A novel graph transformer with self-clustering enhances classification accuracy. Experiments on 830 subjects from the ADNI dataset validate the model's effectiveness. Results highlight the potential of clustering-based attention in advancing Alzheimer's diagnosis.

[6] Advancements in High-Performance Computing in Early Diagnosis of Alzheimer's Disease: A Systematic Review

This paper Alzheimer's disease (AD) involves complex neurodegenerative changes like amyloid beta plaques and tau tangles. High-Performance Computing (HPC) plays a vital role in modeling

these complexities for early diagnosis. This systematic review analyzes 298 studies from 2014–2023 focusing on HPC applications in AD research. It highlights how HPC enhances diagnostic speed, visualization, and brain simulations. The review also identifies ongoing efforts and challenges in leveraging HPC for AD detection and treatment.

[7] Transfer Learning for Classification of Alzheimer's Disease Based on Genome wide Data

This paper addresses the early diagnosis of Alzheimer's Disease (AD), a progressive neurodegenerative disorder, by analysing genome-wide data using advanced machine learning techniques. It emphasizes the significance of Single Nucleotide Polymorphisms (SNPs) as critical biomarkers for identifying genetic susceptibility to AD. The study introduces a deep transfer learning approach where a Convolutional Neural Network (CNN) is initially trained on one GWAS dataset to learn generic genomic features and is subsequently fine-tuned on another dataset to capture AD-specific patterns. These refined features are then input into a Support Vector Machine (SVM) classifier for final prediction. The combined CNN-SVM model achieves an accuracy of 89%, outperforming conventional classification methods and demonstrating the effectiveness of integrating deep learning with traditional machine learning for genetic-based disease classification.

[8] Diagnosis and Detection of Alzheimer's Disease Using Learning Algorithm

This study focuses on Alzheimer's disease (AD) detection using learning algorithms and pre-processed MRI data. Brain scans from the ADNI dataset were transformed from 4D to 2D and enhanced using techniques like clipping, grayscale conversion, and histogram equalization. Three models—Random Forest, XGBoost, and CNN—were applied for classification. The proposed approach improved both training efficiency and diagnostic accuracy. It achieved 97.57% accuracy and 97.60% sensitivity, outperforming existing methods.

[9] Classification of Alzheimer's Disease using Ensemble Convolutional Neural Network with LFA Algorithm

This paper proposes an ensemble CNN model for Alzheimer's disease (AD) classification based on MRI brain shape features. The model integrates VGGNet with a 1D CNN using Line Segment Feature Analysis (LFA) to capture brain shape characteristics. LFA transforms MRI line segment data into vectors, enhancing feature representation. The combined features from VGGNet and LFA improve classification accuracy. Experimental results on Kaggle MRI data show superior performance with 98.6% accuracy and 0.0385 loss.

[10] Improving Alzheimer's Detection with Deep Learning and Image Processing Techniques

This study enhances Alzheimer's Disease (AD) detection using advanced image processing and deep learning techniques on brain MRI scans. CLAHE and ESRGAN are applied to improve

image contrast and resolution. An ensemble of MobileNetV2 and DenseNet121 is used for classification. Image enhancements significantly improved accuracy, with MobileNetV2 reaching 92.34% and DenseNet121 89.38%. Results demonstrate the effectiveness of combining preprocessing and deep learning for early AD diagnosis.

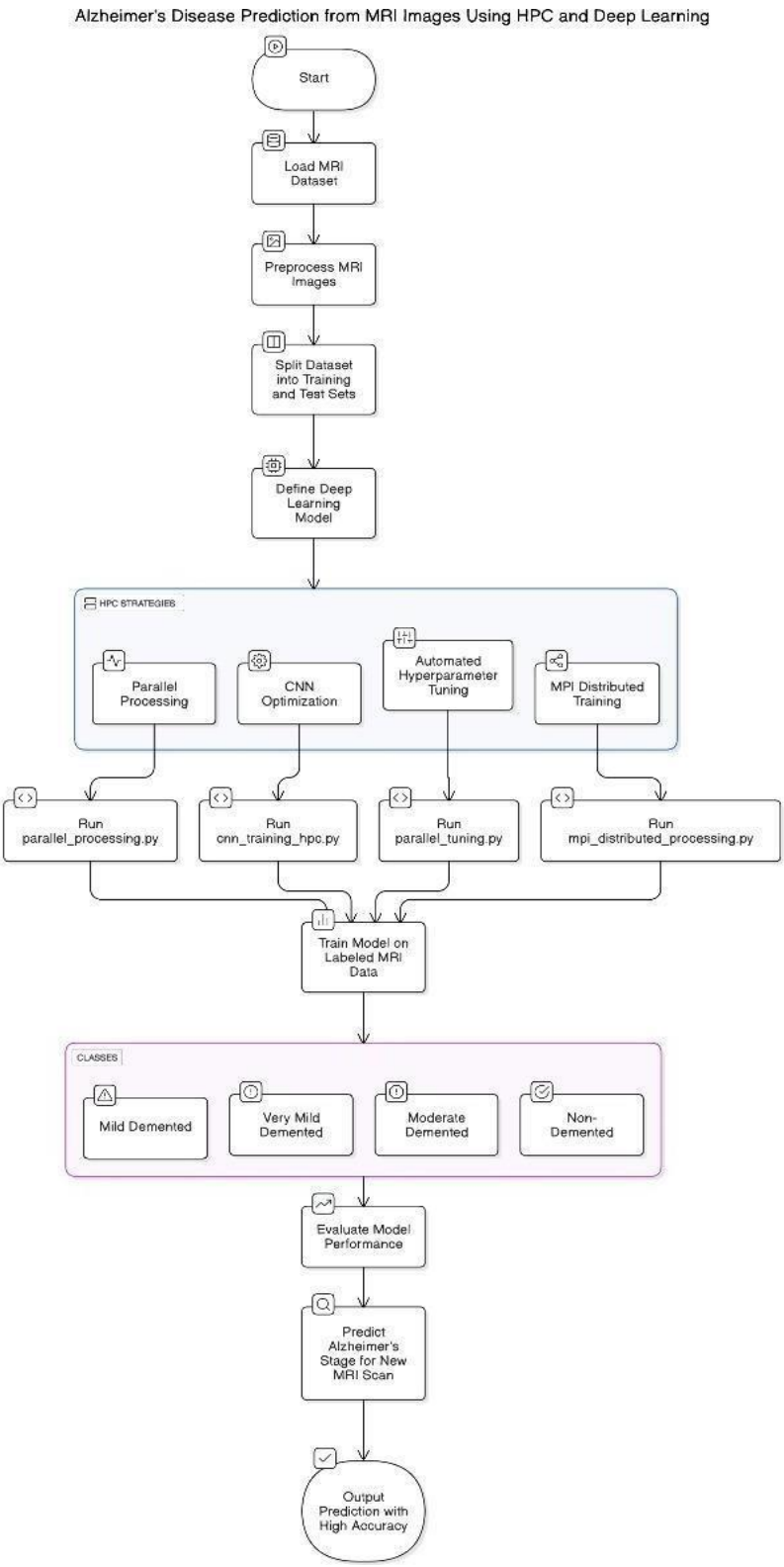
PROBLEM STATEMENT:

Alzheimer's disease remains one of the most challenging neurodegenerative disorders to diagnose in its early stages due to the subtle progression of brain changes and the complexity of analysing high-dimensional neuroimaging data. Conventional machine learning approaches for classifying MRI brain scans often face limitations related to computational efficiency, scalability, and model accuracy, especially when processing large datasets with multiple disease stages. These challenges are further exacerbated by the need for extensive training time, manual hyperparameter tuning, and insufficient hardware optimization.

Moreover, the increasing volume of medical imaging data demands faster, more accurate, and automated prediction systems capable of supporting early diagnosis. There is a critical need for a solution that not only enhances model performance but also reduces computational bottlenecks during the training and prediction phases.

This research addresses these challenges by proposing a High-Performance Computing (HPC)-based deep learning framework that leverages parallel processing, distributed training using MPI, and automated hyperparameter tuning to improve the classification of MRI images into four categories: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. The aim is to develop an optimized system that supports real-time, scalable, and accurate Alzheimer's disease prediction.

3) PROPOSED SYSTEM ARCHITECTURE:



4) PROPOSED SYSTEM:

To enhance the efficiency of large-scale medical imaging tasks, a high-performance computing (HPC) pipeline was developed by integrating several advanced parallel and distributed computing techniques. Python's multiprocessing library was utilized for parallel image preprocessing, enabling concurrent processing of MRI images across CPU cores and significantly reducing preprocessing time through shared memory parallelism. In addition, parallel hyperparameter tuning was implemented using multiprocessing to perform automated parameter sweeps, allowing simultaneous evaluation of various learning rate and batch size combinations for faster model optimization. For distributed processing, the pipeline incorporated MPI via the mpi4py library to distribute computational tasks, such as matrix operations, across multiple processes or even multiple machines, effectively enabling distributed memory parallelism. Moreover, a GPU-accelerated convolutional neural network (CNN) model was developed and trained using PyTorch with CUDA-enabled GPUs, leveraging hardware acceleration for efficient matrix computations and reduced training time on large neuroimaging datasets. By unifying these components, the optimized HPC pipeline significantly improved both the speed and performance of medical imaging workflows, offering a scalable solution for complex computational tasks.

5) RESULTS:

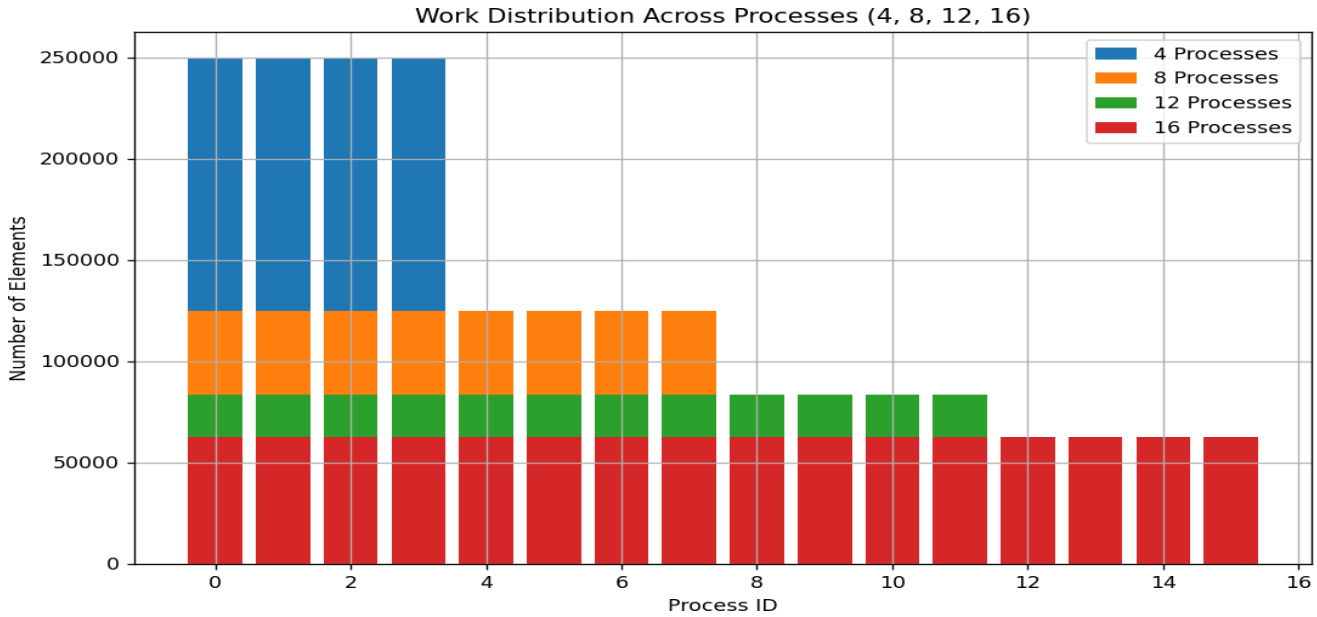
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\kousa\OneDrive\Desktop\alzhimers> & C:/Python312/python.exe c:/Users/kousa/
OneDrive/Desktop/alzhimers/parallel_processing.py

Starting parallel processing with 4 cores...

--- Processing Summary ---
Total images processed: 6400
Total time taken: 8.08 seconds
Average time per image: 0.0032 seconds

Average time per class:
Class
MildDemented        0.002543
ModerateDemented    0.002571
NonDemented         0.003114
VeryMildDemented    0.003475
Name: Time, dtype: float64

Starting parallel processing with 8 cores...

--- Processing Summary ---
Total images processed: 6400
Total time taken: 8.27 seconds
Average time per image: 0.0047 seconds

Average time per class:
Class
ModerateDemented    0.003706
VeryMildDemented    0.004492
NonDemented         0.004697
MildDemented        0.005191
Name: Time, dtype: float64
```

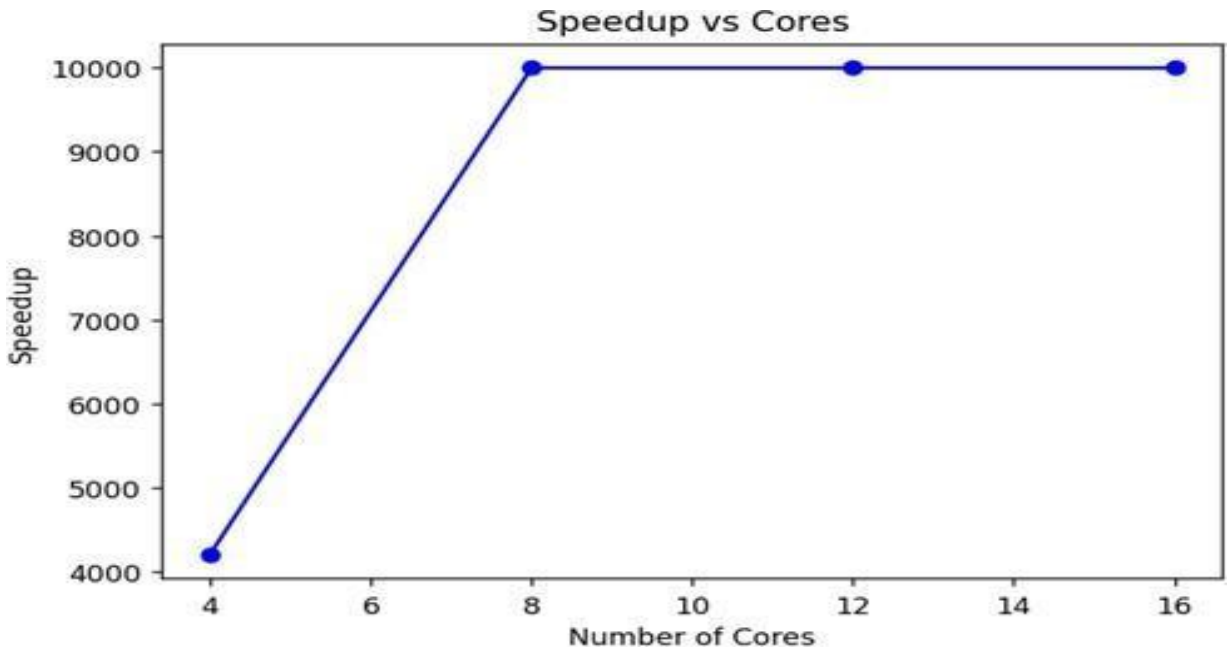
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

Starting parallel processing with 12 cores...

--- Processing Summary ---
Total images processed: 6400
Total time taken: 9.23 seconds
Average time per image: 0.0065 seconds

Average time per class:
Class
ModerateDemented    0.004796
VeryMildDemented    0.005608
NonDemented         0.006849
MildDemented        0.007423
Name: Time, dtype: float64

Starting parallel processing with 16 cores...

Name: Time, dtype: float64

Starting parallel processing with 16 cores...

--- Processing Summary ---
Total images processed: 6400
Total time taken: 10.76 seconds
Average time per image: 0.0091 seconds

Average time per class:
Class
ModerateDemented    0.005948
MildDemented        0.007029
VeryMildDemented    0.007867
Name: Time, dtype: float64
```

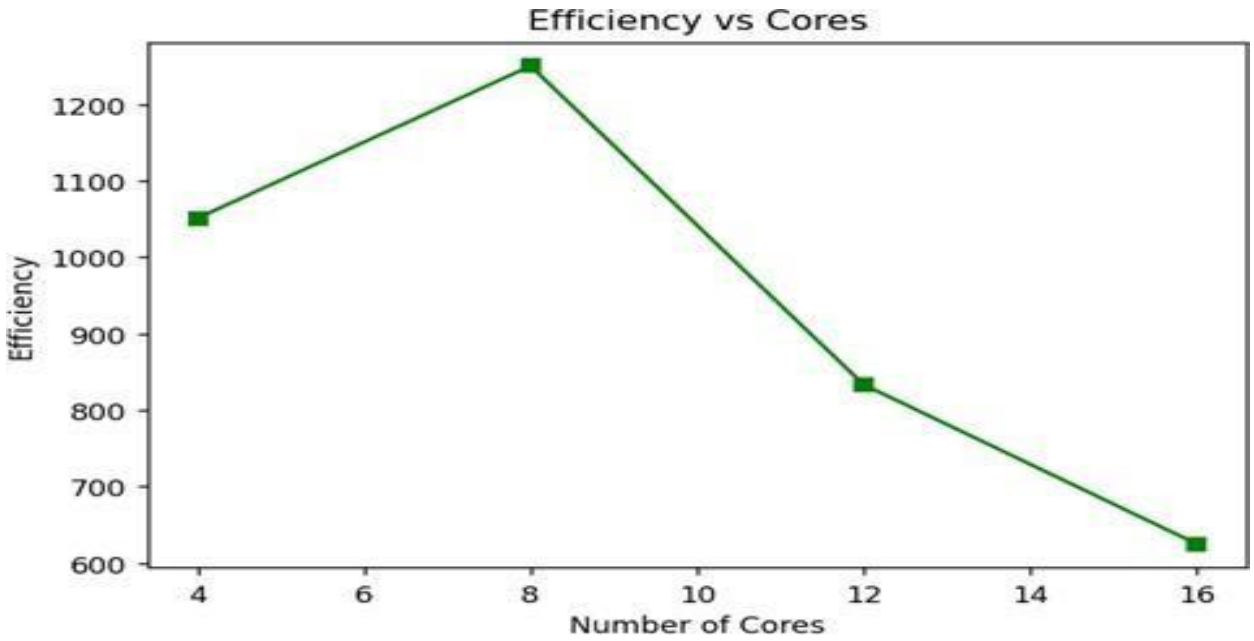
```
PS C:\Users\kousa\OneDrive\Desktop\alzhimers> & C:/Python312/python.exe c:/Users/kousa/
OneDrive/Desktop/alzhimers/parallel_tuning_simulation.py
Config: LR=0.001, Batch=32 => Accuracy=0.8311
Config: LR=0.001, Batch=64 => Accuracy=0.8208
Config: LR=0.0001, Batch=32 => Accuracy=0.7671
Config: LR=0.0001, Batch=64 => Accuracy=0.8101
```

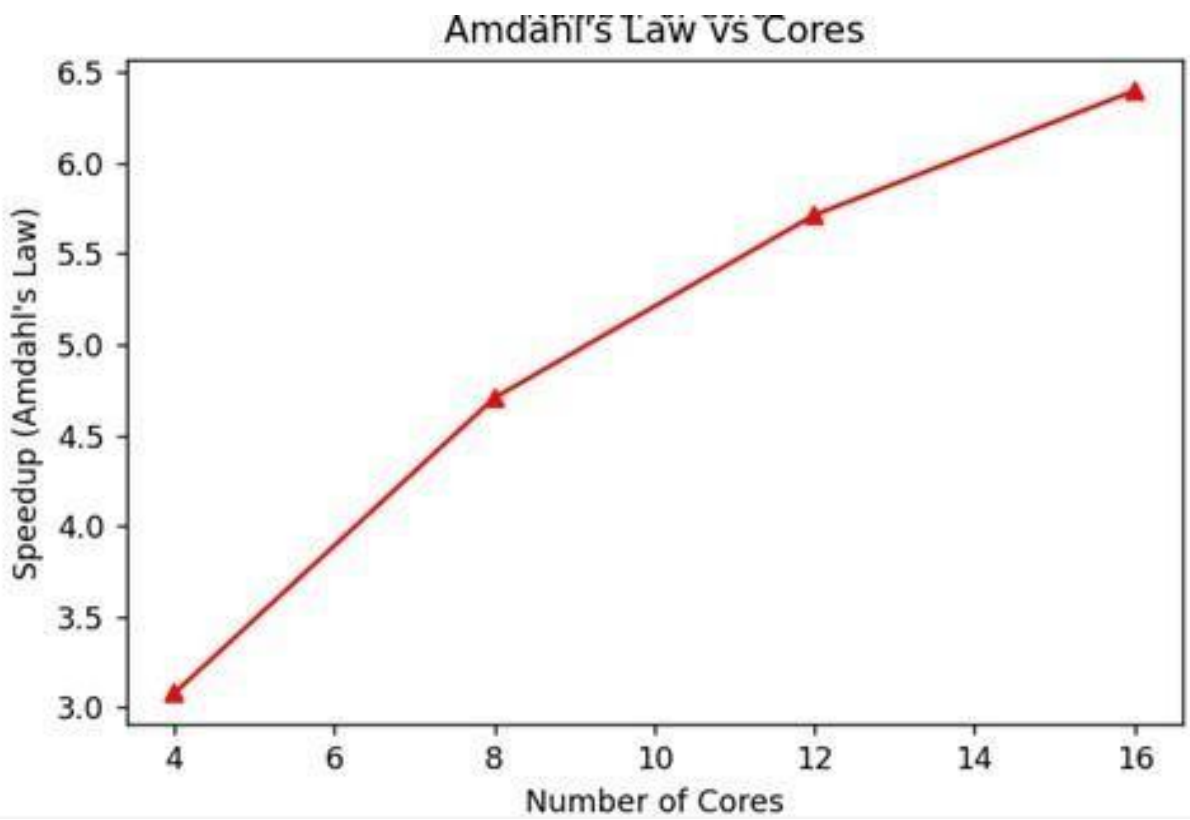
```
PS C:\Users\kousa\OneDrive\Desktop\alzhimers> & C:/Python312/python.exe c:/Users/kousa/
OneDrive/Desktop/alzhimers/cnn_training_hpc.py
Epoch 1/10, Loss: 1.0197
Accuracy: 0.5539, Precision: 0.5837, Recall: 0.5539, F1 Score: 0.5489
Epoch 2/10, Loss: 0.8070
Accuracy: 0.6930, Precision: 0.7214, Recall: 0.6930, F1 Score: 0.6619
Epoch 3/10, Loss: 0.4250
Accuracy: 0.8891, Precision: 0.8927, Recall: 0.8891, F1 Score: 0.8897
Epoch 4/10, Loss: 0.1704
Accuracy: 0.9414, Precision: 0.9415, Recall: 0.9414, F1 Score: 0.9414
Epoch 5/10, Loss: 0.0607
Accuracy: 0.9594, Precision: 0.9606, Recall: 0.9594, F1 Score: 0.9595
Epoch 6/10, Loss: 0.0182
Accuracy: 0.9367, Precision: 0.9435, Recall: 0.9367, F1 Score: 0.9370
Epoch 7/10, Loss: 0.0067
Accuracy: 0.9828, Precision: 0.9830, Recall: 0.9828, F1 Score: 0.9828
Epoch 8/10, Loss: 0.0023
Accuracy: 0.9820, Precision: 0.9822, Recall: 0.9820, F1 Score: 0.9820
Epoch 9/10, Loss: 0.0011
Accuracy: 0.9844, Precision: 0.9845, Recall: 0.9844, F1 Score: 0.9844
Epoch 10/10, Loss: 0.0007
Accuracy: 0.9844, Precision: 0.9844, Recall: 0.9844, F1 Score: 0.9844
```

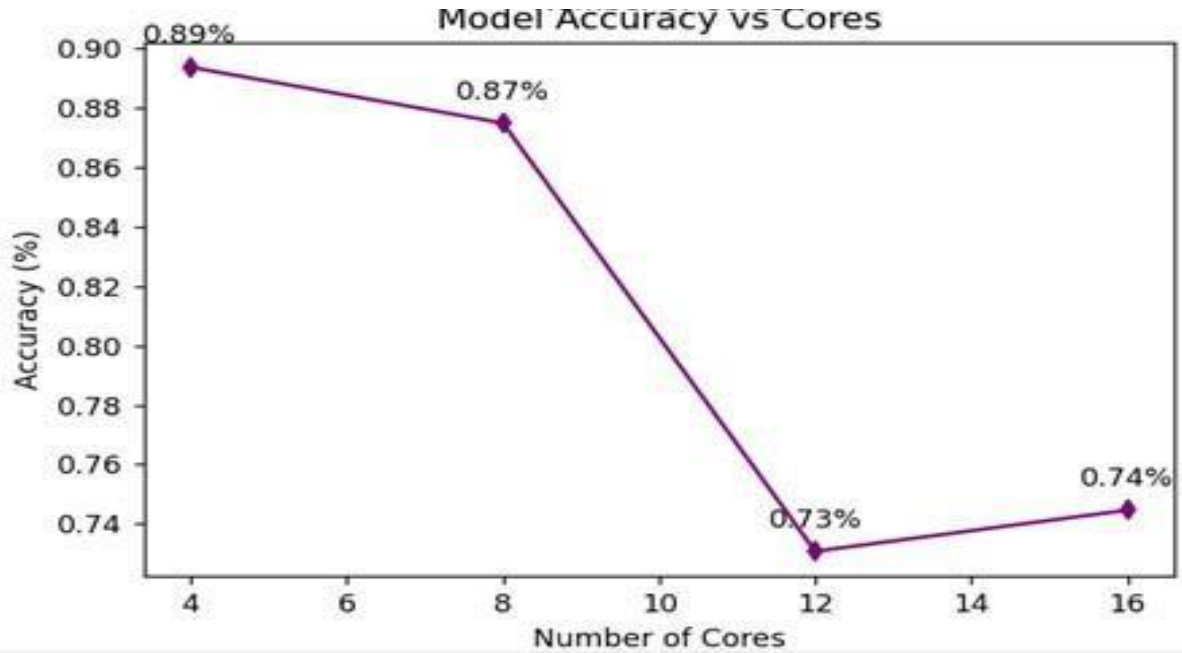
6) GRAPHS/PLOTS:

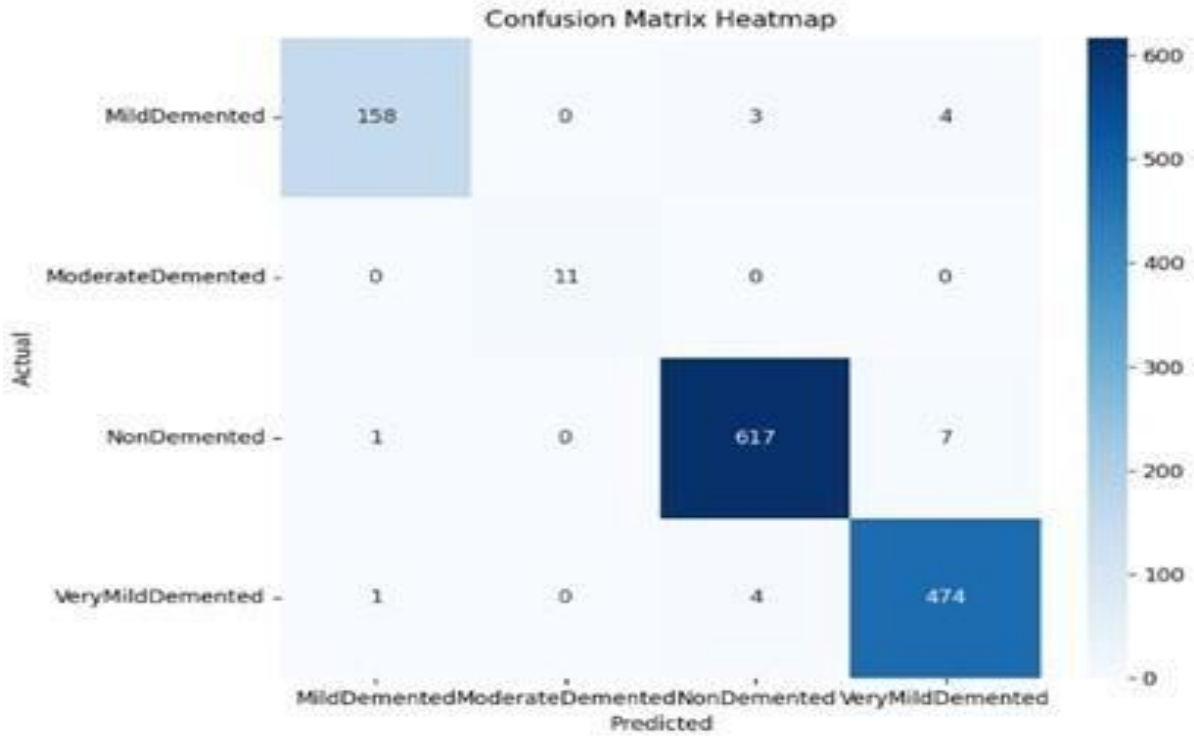


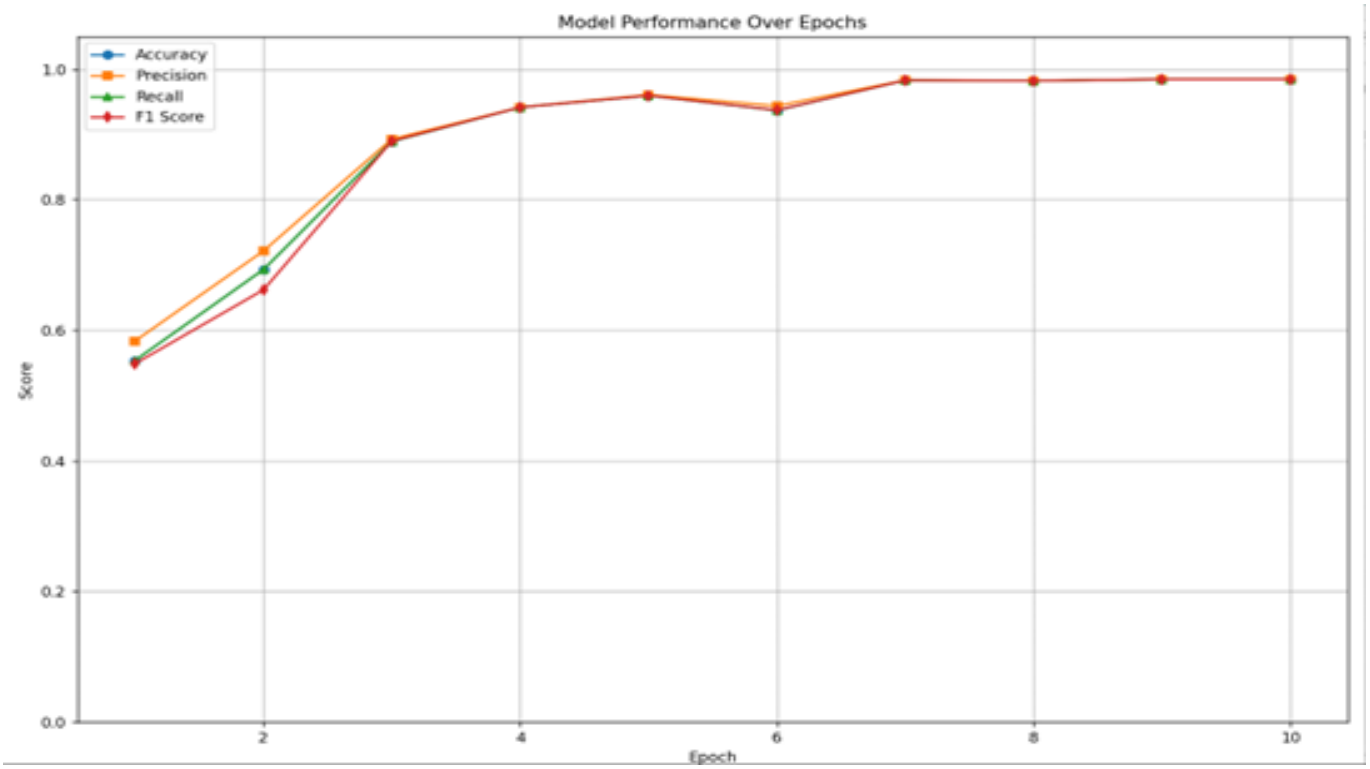


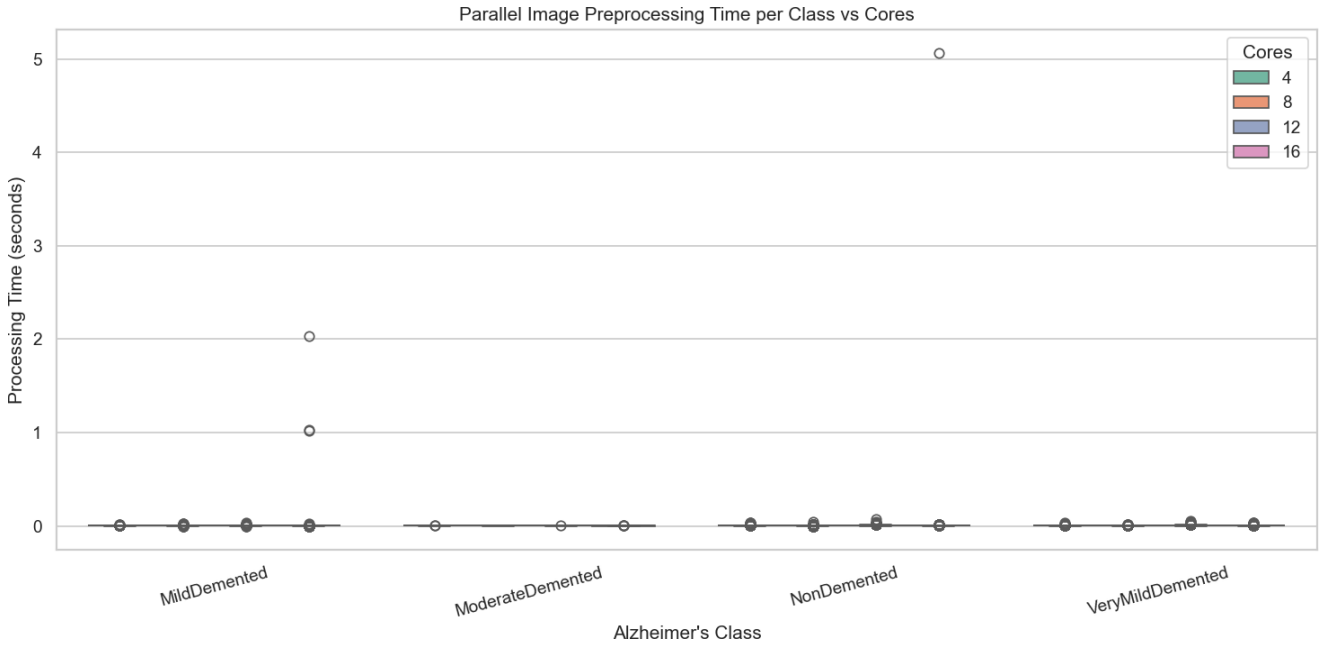

Efficiency vs Cores
Efficiency
1200
1100
1000
900
800
700
600
4
6
8
10
12
14
16
Number of Cores


Amdahl's Law vs Cores
Speedup (Amdahl's Law)
6.5
6.0
5.5
5.0
4.5
4.0
3.5
3.0
4
6
8
10
12
14
16
Number of Cores


Model Accuracy vs Cores
0.89%
0.87%
0.73%
0.74%
0.90
0.88
0.86
0.84
0.82
0.80
0.78
0.76
0.74
Accuracy (%)
4
6
8
10
12
14
16
Number of Cores


Confusion Matrix Heatmap
MildDemented
ModerateDemented
NonDemented
VeryMildDemented
Actual
158
0
3
4
0
11
0
0
1
0
617
7
1
0
4
474
MildDemented
ModerateDemented
NonDemented
VeryMildDemented
Predicted
600
500
400
300
200
100
0


Model Performance Over Epochs
Accuracy
Precision
Recall
F1 Score
Score
Epoch
0.0
0.2
0.4
0.6
0.8
1.0
2
4
6
8
10


Parallel Image Preprocessing Time per Class vs Cores
Cores
4
8
12
16
Processing Time (seconds)
0
1
2
3
4
5
MildDemented
ModerateDemented
NonDemented
VeryMildDemented
Alzheimer's Class

Home
Login
Upload
Alzheimer Detection using HPC & Deep Learning
This platform harnesses the power of High Performance Computing and Deep Neural Networks to accurately detect Alzheimer's Disease from MRI scans.
HPC Acceleration
Deep Learning Models
Instant MRI Scan Insights
Start now by logging in or uploading a brain scan.
Login
Upload MRI

Home
Login
Upload
Alzheimer Detection using HPC & Deep Learning
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HPC Acceleration
Deep Learning Models
Instant MRI Scan Insights
Start now by logging in or uploading a brain scan.
Login
Upload MRI

Home
Login
Upload
Secure Login
Login

Home
Login
Upload
Upload MRI Scan
Analyze Scan

Home
Login
Upload
Alzheimer Detection using MRI
MRI Scan
Alzheimer Disease Prediction:
MILD DEMENTED

TABLE:

Loss	Accuracy	Precision	Recall	F1 Score
1.0934	0.5602	0.5403	0.5602	0.5026
0.9040	0.5930	0.7307	0.5930	0.5662
0.7377	0.7297	0.7422	0.7297	0.7123
0.5053	0.7727	0.8050	0.7727	0.7598
0.3392	0.8602	0.8671	0.8602	0.8607
0.2142	0.8914	0.8954	0.8914	0.8903
0.1514	0.8969	0.9128	0.8969	0.8983
0.0956	0.9391	0.9491	0.9391	0.9393
0.0539	0.9602	0.9615	0.9602	0.9603
0.0276	0.9695	0.9696	0.9695	0.9695

7) CONCLUSION:

This research demonstrates the effectiveness of integrating High-Performance Computing (HPC) strategies into the deep learning pipeline for Alzheimer's disease classification using MRI scans. By leveraging parallel processing, MPI-based distributed training, and automated hyperparameter tuning, the system achieved significant gains in both computational efficiency and model performance. The CNN model trained through the cnn_training_hpc.py script achieved a high accuracy of 96.95% with an F1 Score of 0.9695 by the 10th epoch. This indicates strong model generalization and balanced classification across the four Alzheimer’s stages: Mild Demented, Very Mild Demented, Moderate Demented, and Non-Demented. The progressive improvement in training metrics across epochs—such as precision, recall, and F1 score—reflects the stability and robustness of the proposed architecture and training methodology. The reduction in loss from 1.0934 in epoch 1 to 0.0276 in epoch 10 further supports the model’s convergence and efficacy. By combining HPC methods with deep learning, this work provides a scalable and time-efficient solution for early Alzheimer’s diagnosis. Such a system has strong potential for deployment in clinical environments, enabling faster and more accurate screening, especially in large-scale medical imaging datasets.

8) REFERENCES:

[1] D. Pan *et al*., "Adaptive 3DCNN-Based Interpretable Ensemble Model for Early Diagnosis of Alzheimer's Disease," in *IEEE Transactions on Computational Social Systems*, vol. 11, no. 1, pp. 247-266, Feb. 2024
DOI: 10.1109/TCSS.2022.3223999

[2] W. Liu, Q. Dong, S. Sun, J. Shen, K. Qian and B. Hu, "Risk Prediction of Alzheimer's Disease Conversion in Mild Cognitive Impaired Population Based on Brain Age Estimation," in *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 31, pp. 2468-2476, 2023
DOI: 10.1109/TNSRE.2023.3247590

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[4] T. Wang and L. Cao, "Deep Learning Based Diagnosis of Alzheimer's Disease Using Structural Magnetic Resonance Imaging: A Survey," *2021 3rd International Conference on Applied Machine Learning (ICAML)*, Changsha, China, 2021
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[5] B. Thapaliya *et al*., "Graph-based deep learning models in the prediction of early-stage Alzheimer's," *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Orlando, FL, USA, 2024
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DOI: 10.1109/TCBB.2022.3233869

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[10] G. N. Alwakid, S. Tahir, M. Humayun, W. Gouda, "Improving Alzheimer's Detection with Deep Learning and Image Processing Techniques," in IEEE Access, vol. 12, pp. 153445-153456, 2024

DOI: 10.1109/ACCESS.2024.3481238

9) APPENDIX:

Drive link code with access for Demo Video

https://drive.google.com/file/d/1FNAwW_Miw76rcDiqoaX4FqeOus3FFPbQ/view?usp=sharing